



ARGENTINA VS SPAIN



MATCH WINNER PREDICTION

```
1 # Football Match Winner Prediction Program
2 import numpy as np
3 import pandas as pd
4
5 def get_team_data():
6     data = {
7         'Argentina': {'attack': 8.8, 'midfield': 8.5,
8                       'defense': 8.3, 'form': 8.6},
9         'Spain': {'attack': 8.2, 'midfield': 8.7,
10                  'defense': 8.0, 'form': 8.3}
11     }
12     return data
13
14 # Calculate weighted score
15 def calculate_score(team):
16     weights = {'attack': 0.35, 'midfield': 0.30,
17               'defense': 0.20, 'form': 0.15}
18     score = 0
19     for key in weights:
20         score += team[key] * weights[key]
21     return round(score, 2)
22
23 # Prediction Logic
24 def predict_winner(data):
25     arg_score = calculate_score(data['Argentina'])
26     spa_score = calculate_score(data['Spain'])
27     diff = round(arg_score - spa_score, 2)
28     if diff > 0:
29         winner = "Argentina"
30     elif diff < 0:
31         winner = "Spain"
32     else:
33         winner = "Draw"
34     total = arg_score + spa_score
35     prob_arg = round((arg_score / total) * 100, 2)
36     prob_spa = round((spa_score / total) * 100, 2)
37     return {
38         'arg_score': arg_score,
39         'spa_score': spa_score,
40         'diff': diff,
41         'winner': winner,
42         'prob_arg': prob_arg,
43         'prob_spa': prob_spa
44     }
45
46 # Run Program
47 data = get_team_data()
48 result = predict_winner(data)
49 print("===== PREDICTION RESULT =====")
50 print(f"Argentina Score : {result['arg_score']}")
51 print(f"Spain Score : {result['spa_score']}")
52 print(f"Difference : {result['diff']}")
53 print(f"Winner : {result['winner']}")
54 print(f"Win Probability -> Argentina: {result['prob_arg']}% | Spain: {result['prob_spa']}%")
```

PREDICTION RESULT

Argentina Score : 8.48
Spain Score : 8.34
Difference : +0.14
Winner : Argentina
Win Probability -> Argentina: 50.44% | Spain: 49.56%

WIN PROBABILITY

50.44%
ARGENTINA



49.56%
SPAIN



KEY FACTORS (OUT OF 10)

⚡ ATTACK	8.8	VS	8.2
🎯 MIDFIELD	8.5	VS	8.7
🛡 DEFENSE	8.3	VS	8.0
📈 FORM	8.6	VS	8.3

DATA SOURCE

- Recent Performance (Last 10 Matches)
- Team Statistics & Ratings
- Attack, Midfield, Defense & Form Index
- ELO Rating & Head-to-Head Records



FINAL PREDICTION

ARGENTINA TO WIN

Very close match! But Argentina has a slight edge.



PYTHON IN 30 SECONDS #022

deque.rotate()

Rotate elements efficiently without manually shifting a list.

```
1  from collections import deque
2  services = deque(["Auth", "Billing", "Search"])
3  services.rotate(1)
4
5  print(services)
```

Output

```
deque(['Search', 'Auth', 'Billing'])
```



Quick Tip

`rotate()` is ideal for circular queues, schedulers and round-robin algorithms because it reorders elements efficiently without manually moving list items.

CODEXIMMORTAL

AGENTIC AI WAR LAB

SYSTEM OVERVIEW



NETWORK STATUS
ONLINE

NODE INTEGRITY

98.7%

THREAT INDEX
LDW



SIGNAL ANALYSIS



LATENCY
12.7 ms



SENTINEL



STRATEGIST



ARCHIVIST



EXECUTOR

PQC PROTOCOL

CRYSTALS-DILITHIUM
FALCON
SPHINCS+
ACTIVE

ENTROPY HEALTH



99.3 %

LEDGER STATUS

DISTRIBUTED
VERIFIED



AI COHERENCE



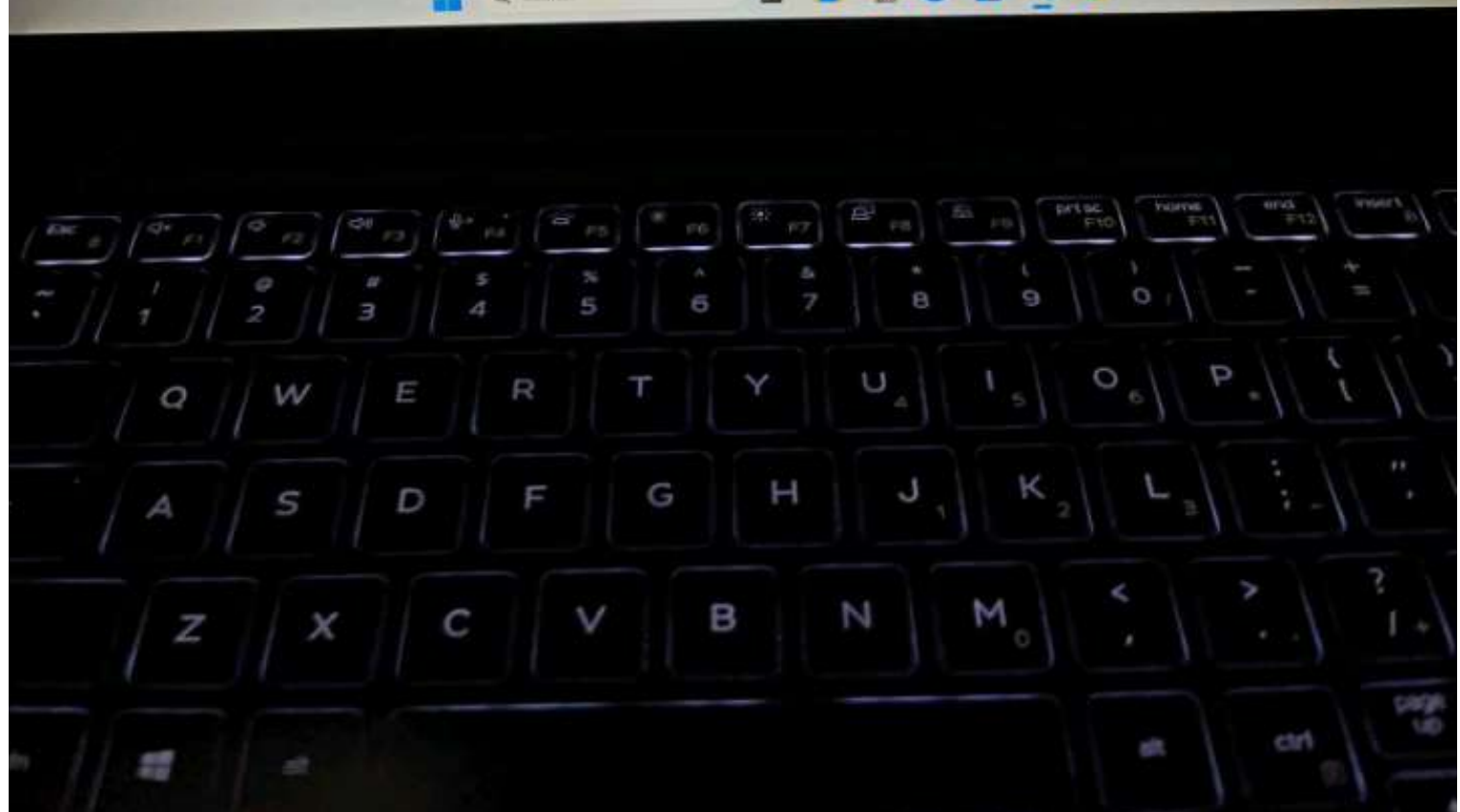
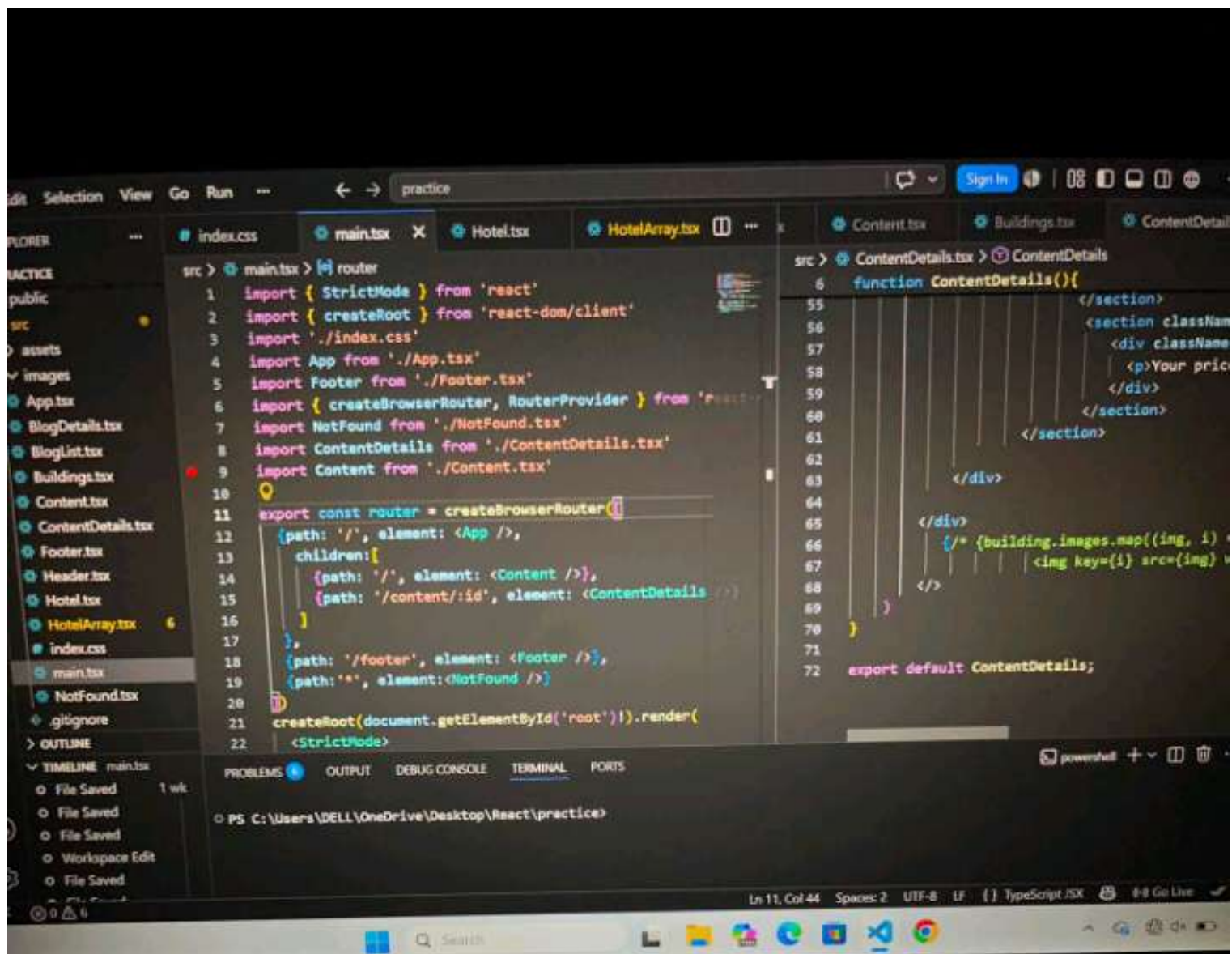
96.4%

GLOBAL NODES



247
ONLINE

PRIVATE AI • PQC TRUST • HUMAN GATE



JS

script.js

```
1 // Expense Tracker - JavaScript
2 document.addEventListener("DOMContentLoaded", () => {
3   // Elements
4   const descriptionInput = document.getElementById("description");
5   const amountInput = document.getElementById("amount");
6   const categoryInput = document.getElementById("category");
7   const addButton = document.getElementById("add-expense");
8   const totalAmountEl = document.getElementById("total-amount");
9   const monthlyAmountEl = document.getElementById("Monthly-amount");
10  const tbody = document.getElementById("expense-body");
11
12  let expenses = [];
13
14  // Add Expense
15  addButton.addEventListener("click", addExpense);
16
17  function addExpense() {
18    const description = descriptionInput.value.trim();
19    const amount = parseFloat(amountInput.value);
20    const category = categoryInput.value;
21
22    if (!description || isNaN(amount) || amount <= 0) {
23      alert("Please fill in all fields correctly.");
24      return;
25    }
26
27    const expense = {
28      id: Date.now(),
29      description,
30      amount,
31      category,
32      date: new Date().toLocaleDateString()
33    };
34
35    expenses.push(expense); // Add to list
36    renderExpenses();
37    updateSummary();
38    clearInputs();
39  }
40
41  // Render Expenses Table
42  function renderExpenses() {
43    tbody.innerHTML = "";
44    expenses.forEach(expense => {
45      const row = document.createElement("tr");
46      row.innerHTML = `
47        <td>${expense.description}</td>
48        <td>${expense.category}</td>
49        <td>${expense.amount.toFixed(2)}</td>
50        <td>${expense.date}</td>
51        <td>
52          <button class="delete-btn" data-id="${expense.id}">✖</button>
53        </td>
54      `;
55      tbody.appendChild(row);
56    });
57  }
```

Expense Tracker

Track your expenses. Build a better tomorrow.

Description

e.g., Groceries

Amount

0.00

Category

Select Category

+ Add Expense



Total Expenses

\$668.50



This Month

\$528.50

Recent Expenses

[View All](#)

Description	Category	Amount	Date	Action
Groceries	Food	\$45.50	May 18, 2024	
Bus Pass	Transport	\$60.00	May 17, 2024	
Electricity Bill	Utilities	\$120.00	May 16, 2024	
Headphones	Entertainment	\$80.00	May 14, 2024	
Movie Ticket	Entertainment	\$15.00	May 14, 2024	
Dinner	Food	\$48.00	May 13, 2024	

Showing 1 to 6 of 6 expenses



Follow for More

Code: Python

```
a = 8
```

```
b = 3
```

```
result = a + b * 2
```

```
print(result)
```

What is the Output?

A). 14

B). 16

C). 22

D). Compile Time Error

@Bhanu Srivastava 07

COMMENT YOUR ANSWER

HTML LINKS



CS ACADEMY
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HTML links are used to connect one webpage to another webpage, a file, an email address, or any location on the web.

1. What is a Link?

- A link is a clickable element that takes you to another page or resource.
- Links are created using the `<a>` tag.



2. Basic Syntax

```
<a href="URL">Link Text</a>
```

Example :

```
<a href="https://www.google.com">Visit Google</a>
```



3. Types of Links

- Link to another webpage

```
<a href="about.html">About Us</a>
```

Opens another page on the same website.

- Link to an external website

```
<a href="https://www.wikipedia.org">Wikipedia</a>
```

Opens an external website.



- Link to an email address

```
<a href="mailto:info@csacademy.com">Email Us</a>
```

Opens the default email application.



- Link to a file (for download)

```
<a href="notes.pdf" download>Download Notes</a>
```

Downloads the file when clicked.



- Link to a section on the same page

```
<a href="#contact">Go to Contact Section</a>
```

Takes you to a specific section on the same page.



4. Important Attributes of <a> Tag

- **href** : Specifies the destination URL.
- **target** : Specifies where to open the linked document.
- **title** : Provides extra information about the link.
- **download** : Forces download of the linked resource.

Example with Attributes :

```
<a href="https://www.csacademy.com" target="_blank" title="Visit CS Academy" download>CS Academy</a>
```

5. Summary

- Links help users navigate between pages and resources.
- Use the `<a>` tag with **href** to create a link.
- Use **target="_blank"** to open links in a new tab.



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WHAT PEOPLE **THINK** A
NETWORK ENGINEER DOES

VS

WHAT A NETWORK ENGINEER
REALLY DOES



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NETWORKING CONCEPTS

IPv4 NETWORK ADDRESSING

≧ **8 MCQs** ≦



1

What is the standard subnet mask for an IPv4 Class A network?

- A) 255.255.0.0 B) 255.0.0.0 C) 255.255.255.0 D) 255.255.255.255

2

Which class of IPv4 addresses does Automatic Private IP Addressing (APIPA) allocate to Dynamic Host Configuration Protocol (DHCP) clients when they are unable to reach a DHCP server?

- A) Class A B) Class B C) Class C D) Class D

3

What is the valid range of host addresses that can be utilized for workstations if the IPv4 network address is 192.168.1.32/27?

- A) 192.168.1.33 through 192.168.1.63 B) 192.168.1.34 through 192.168.1.62
C) 192.168.1.34 through 192.168.1.63 D) 192.168.1.33 through 192.168.1.62

4

Linda has been given the IPv4 network address 172.21.0.0/22 to set up a new department network in her company. How many host addresses are available to her?

- A) 1,022 B) 510 C) 512 D) 1,024

5

Where is the address 127.0.0.1 found in the IPv4 class?

- A) Class A B) Class B C) Class C D) None of the options

6

What class does the IPv4 address 190.126.14.251 belong to?

- A) Class A B) Class B C) Class C D) Class D E) Class E

7

What is the equivalent decimal value of the binary mask 11111111.11111111.11100000.00000000?

- A) 255.255.224.0 B) 255.255.240.0 C) 255.255.248.0 D) 255.255.252.0

8

Which service from the options allows computers on a private IPv4 network to connect to the Internet with a registered IP address?

- A) DHCP B) DNS C) NAT D) NTP



STUDY SMART, SCORE HIGH!

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- 🔗 Share with your friends
- 👤 Follow for more MCQs & Notes



RJ45 Ethernet Connector (8P8C)

RJ45 Connector (8P8C)

Gold-Plated Contacts

Transparent Housing

Locking Clip

Internal Wire Channels

Copper Conductors

Cable Entry

Strain Relief Area

What is an RJ45 Connector?

- Standard 8P8C (8 Positions, 8 Contacts) modular connector.
- Commonly used for Ethernet networking.
- Supports LAN communication in homes, offices, enterprise networks, and data centers.
- Compatible with Cat5e, Cat6, Cat6a, Cat7 (RJ45-compatible), and Cat8 Ethernet cabling when used with compatible connectors and networking hardware.

• 8 Contacts

Gold-plated pins for reliable electrical connection.

• Locking Clip

Secures the connector firmly inside an Ethernet port.

• Transparent Housing

Allows visual inspection of wire placement before crimping.

• Used With



Quick Tip

Always verify the wire order before crimping. Once the contacts are crimped, correcting an incorrect wire sequence requires cutting off the connector and installing a new one.

STATEFUL vs STATELESS FIREWALL



Main Difference: Stateless firewall checks each packet individually.
Stateful firewall remembers the connection / session.



1. STATELESS FIREWALL

A stateless firewall only checks the current packet. It does not remember any previous session.



What it checks:

- Source IP
- Destination IP
- Port
- Protocol

Every packet is treated as a new request.

Advantages



- Fast processing
- Less resource usage
- Simple configuration

Limitations



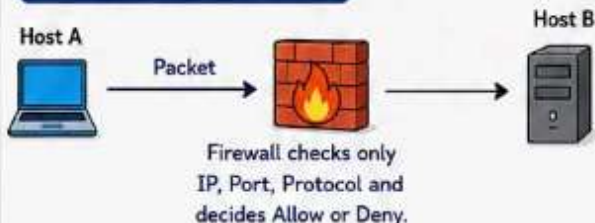
- No session awareness
- Less secure
- Cannot detect advanced attacks properly



Example

ACL on routers is mostly stateless.

How it works (Stateless)



2. STATEFUL FIREWALL

A stateful firewall keeps track of active sessions and connections. It understands the context of traffic.



What it understands:

- Who initiated the connection
- Session state
- Return traffic
- TCP handshake

It stores this information in a State Table.

Advantages



- Better security
- Intelligent traffic inspection
- Detects suspicious sessions
- Allows only legitimate return traffic

Limitations



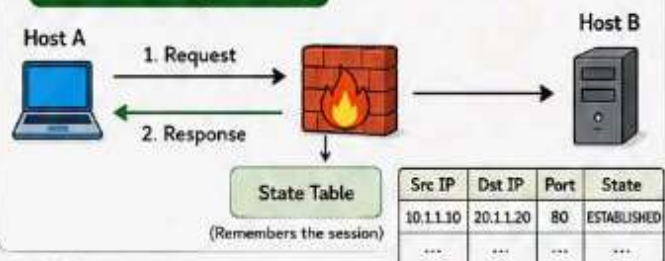
- Uses more CPU / RAM
- Slightly complex



Example

Modern firewalls like Palo Alto, Fortinet, Cisco NGFWs are stateful.

How it works (Stateful)



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- Stateless firewall asks → "Should I allow this packet?"
- Stateful firewall asks → "Does this packet belong to a valid connection?"



KEY DIFFERENCES

Point	Stateless Firewall	Stateful Firewall
1 Checks	Checks packets individually	Tracks complete session / connection
2 Memory	No memory	Maintains state table
3 Speed	Faster	Slightly slower
4 Security	Basic filtering	More secure
5 Inspection	No deep inspection	Deep inspection
6 Intelligence	Less intelligent	Smart traffic handling
7 Example	Router ACL	Palo Alto, Fortinet, Cisco NGFW

NOTE



Stateful firewalls are widely used in modern enterprise networks because they understand traffic, not just packets.



WHAT IS LINK-STATE ROUTING?



Link-State Routing is a routing method where routers build a **complete map** of the network and calculate the **shortest path** to each destination.

HOW IT WORKS

1 Discover neighbors and measure link costs.



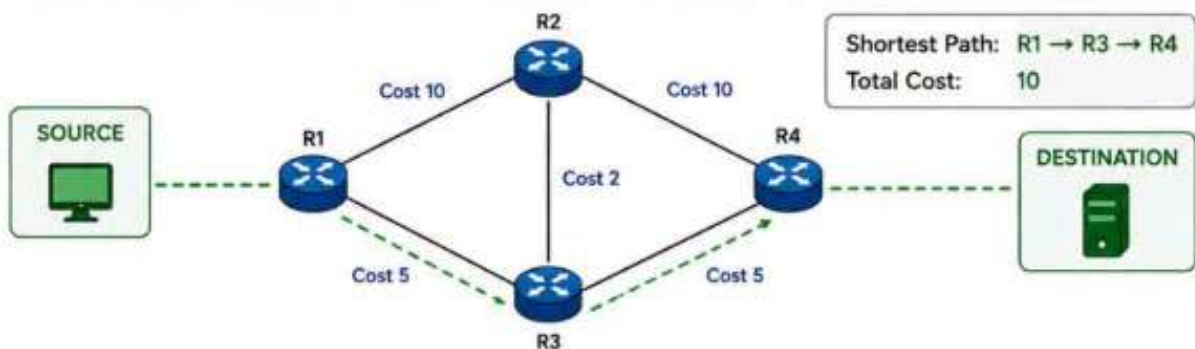
2 Share link-state information with neighbors.



3 Build a complete map (topology) of the network.



4 Calculate the shortest path to each destination.



Fast Convergence



Accurate Routing



Quickly Adapts to Network Changes

Example Protocol:

OSPF



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Next
Generation
Internet!

IPv4 vs IPv6

More Address
More Security
More Future!

THE EVOLUTION OF INTERNET PROTOCOL

IPv4 (Internet Protocol version 4)

★ 32-bit address

Example: 192.168.1.1

★ About 4.3 Billion addresses

★ Dotted Decimal Notation

Example: 192.168.1.1

★ Variable Length Header
(20 to 60 bytes)

★ Supports Broadcast
Example: ARP, DHCP

★ IPSec is Optional

★ NAT is Commonly Used
(Due to limited addresses)

★ Manual or DHCPv4



ADDRESS
LENGTH



ADDRESS
SPACE



NOTATION



HEADER



BROADCAST
vs
MULTICAST



SECURITY



NAT



CONFIGURATION

IPv6 (Internet Protocol version 6)

★ 128-bit address

Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

★ About 340 Undecillion addresses
(340 followed by 36 zeros)

★ Colon Hexadecimal Notation

Example: 2001:0db8:85a3::8a2e:0370:7334

★ Fixed Length Header
(40 bytes)

★ No Broadcast, uses Multicast
Example: Neighbor Discovery

★ IPSec is Built-in (Mandatory Support)

★ NAT is Not Required
(Huge address space)

★ SLAAC and/or DHCPv6



IPv4 – Good, But Running Out of Addresses!

IPv6 – The Future of Internet!

PM NETWORKING

IP PHONE REGISTRATION IN CUCM

End-to-End Signaling & Configuration Flow

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1 PHONE POWER ON

- IP phone boots up



Phone Boot Process

- Power On Self Test
- Load Bootstrap
- DHCP Discover
- TFTP Request (Option 150)

2 DHCP PROCESS



DHCP Server

DHCP Server Response

- IP Address
- Subnet Mask
- Default Gateway
- Option 150 (TFTP Server IP)
- Option 66 (TFTP Server IP)
- Option 42 (NTP Server IP)

- Signaling (SIP)
- TFTP
- Database Replication
- Other / Data Flow

COMMON DHCP OPTIONS

66	TFTP Server IP
150	TFTP Server IP
42	NTP Server IP
3	Default Gateway
6	DNS Server IP
51	Lease Time

3 TFTP CONFIG DOWNLOAD



TFTP Server



Phone Downloads

- Configuration File (SEPxxxx.cnf.xml)
- Firmware File (If required)

- TFTP Port
- UDP 69

4 CONTACTING CUCM (SUBSCRIBER)



Phone Sends SIP Request

- REGISTER to Subscriber node (CallManager)

SIP Ports

- TCP 5060
- TLS 5061

Database Replication (Informix)

- Replication to Publisher
- TCP Port: 3343

5 AUTHENTICATION & DATABASE CHECK

CUCM Performs

- Device Authentication
- Credential Validation
- Device Security Profile Check
- Line / Partition Check
- User Credential Check

6 PHONE REGISTERED

- Phone status: **Registered**
- Ready for calls

Example Phone Status

Status: Registered
IP: 10.10.10.50
Active Load ID: 2
Primary CUCM: 10.10.10.10

Example Phone Status

Status: Registered
IP: 10.10.10.50
Active Load ID: 2
Primary CUCM: 10.10.10.10



IP PHONES

TECHNICAL FLOW SUMMARY

1 POWER ON



2 DHCP



3 TFTP



4 REGISTER



5 AUTHENTICATE



6 REGISTERED



PACKET vs FRAME

Understanding the Difference

OSI LAYERS

- 7 APPLICATION LAYER
- 6 TRANSPORT LAYER
- 5 NETWORK LAYER
- 4 DATA LINK LAYER
- 3 PHYSICAL LAYER

PACKET (Layer 3)

A Packet is a Layer 3 (Network Layer) PDU used for communication across networks. It contains logical addressing information (IP addresses).



ROUTER

IP PACKET (LAYER 3)

IP HEADER	DATA (PAYLOAD)
Source IP 192.168.1.1	HELLO
Destination IP 10.0.0.1	

Packets travel across multiple networks using logical IP addressing.



FRAME (Layer 2)

A Frame is a Layer 2 (Data Link Layer) PDU used for communication within the same local network. It contains MAC addressing information.



SWITCH

ETHERNET FRAME (LAYER 2)

DEST MAC	SRC MAC	TYPE	DATA (PAYLOAD)	FCS
AA:BB:CC:DD:EE:FF	11:22:33:44:55:66	0x0800 (IP)	HELLO	CRC

Frames are used within the same local network using MAC addresses.



OSI LAYERS

- 7 APPLICATION LAYER
- 6 TRANSPORT LAYER
- 5 NETWORK LAYER
- 4 DATA LINK LAYER
- 3 PHYSICAL LAYER

KEY POINTS

- Operates at Layer 3
- Uses IP Addresses
- Used for End-to-End Communication
- Handled by Routers

- Operates at Layer 2
- Uses MAC Addresses
- Used for Local Communication
- Handled by Switches

PACKET vs FRAME - AT A GLANCE

CATEGORY	PACKET (LAYER 3)	FRAME (LAYER 2)
Definition	A Layer 3 PDU used to send data across different networks.	A Layer 2 PDU used to send data within the same local network.
Addressing	Uses IP Addresses (Logical Address)	Uses MAC Addresses (Physical Address)
Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Devices Involved	Routers	Switches / Bridges
Scope	End-to-End Communication	Node-to-Node (within same network)
Example	IP Packet	Ethernet Frame
Purpose	Routing data across multiple networks	Delivering data between devices on the same network
Header Info	IP Header (Source IP, Destination IP, etc.)	MAC Header (Source MAC, Destination MAC, etc.) + FCS for error checking



What is a Server?



A server provides **services** and **resources** to other devices.

HOW IT WORKS



TYPES OF SERVERS



Web Server
Hosts websites and web applications.



File Server
Stores and manages files for users.



Database Server
Manages and provides database access.



Mail Server
Sends, receives, and manages emails.



Application Server
Runs applications and business logic.



Security Server
Provides security services and protection.

KEY FEATURES



Always On
Runs 24/7 to provide continuous services.



Reliable
Designed for high availability.



Powerful
Handles multiple requests efficiently.



Secure
Protects data and resources.



Scalable
Can grow to meet increasing needs.



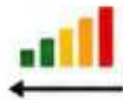
PM Networking



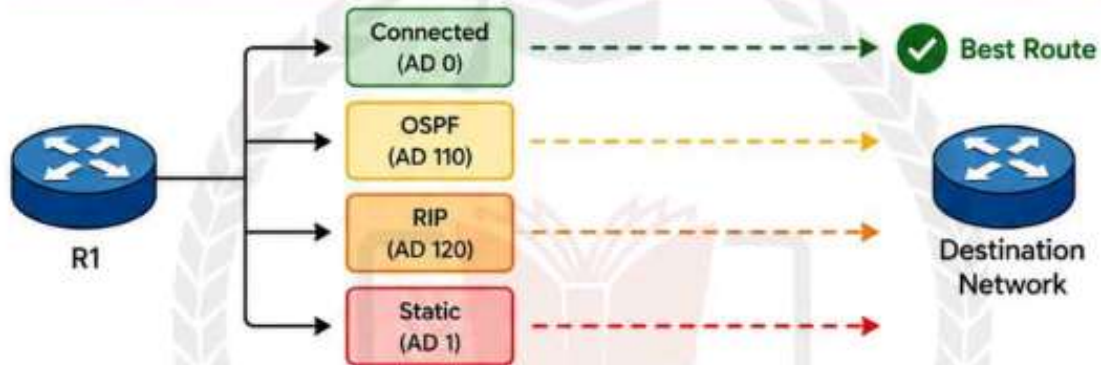
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WHAT IS ADMINISTRATIVE DISTANCE (AD)?



Administrative Distance (AD) measures how **trustworthy** a routing source is. **Lower values** are preferred.



COMMON ADMINISTRATIVE DISTANCES

ROUTING SOURCE	DEFAULT AD VALUE
Connected	0
Static	1
EIGRP (Internal)	90
OSPF	110
RIP	120
EIGRP (External)	170
BGP	200

HOW ROUTERS USE AD



Router learns multiple routes to the same destination.



Router compares the AD of each route.



Route with the **lowest AD** is installed in the routing table.



Lower AD = More trustworthy route

Router prefers the route with the **lowest AD**.



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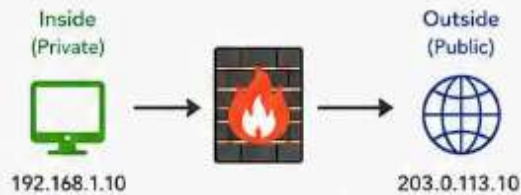
NAT TYPES IN PALO ALTO FIREWALLS

NAT (Network Address Translation) allows private networks to communicate with external networks by **translating IP addresses**.

01

SOURCE NAT (SNAT)

- Modifies the source IP address of outgoing traffic.
- Used when internal users access external networks.



Use Case

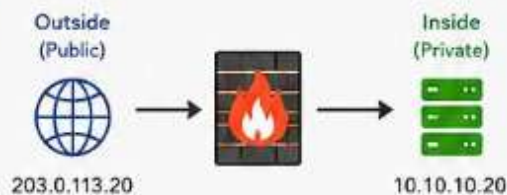


Internal users accessing the Internet.

02

DESTINATION NAT (DNAT)

- Modifies the destination IP address of incoming traffic.
- Used to publish internal servers to the Internet.



Use Case

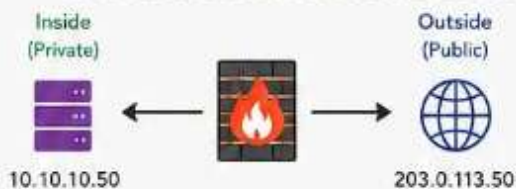


Publishing internal servers (Web, Mail, App) to Internet.

03

STATIC NAT

- Provides a one-to-one mapping between private and public IP addresses.
- Used for servers that need a dedicated public IP.



Use Case

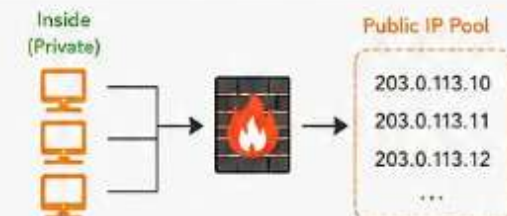


Hosting services that require a dedicated public IP.

04

DYNAMIC IP NAT

- Maps private IP addresses to a pool of public IP addresses dynamically.
- Used when multiple public IPs are available.



Use Case

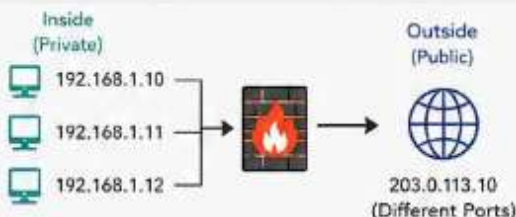


Organizations needing multiple public IPs for outbound traffic.

05

DIPP / PAT (Dynamic IP and Port NAT)

- Multiple internal devices share one public IP using different source ports.
- Most common NAT deployment.



Use Case



Enterprise Internet access for many users with a single public IP.

06

U-TURN NAT (Hairpin NAT)

- Allows internal users to access internal server using its public IP.



Use Case



Internal users accessing company services via public DNS/IP.



TIPS FOR TROUBLESHOOTING NAT

- ✓ Verify security policy matches translated traffic.
- ✓ Check NAT rule order and hit count.
- ✓ Use Traffic Logs to confirm translation.
- ✓ Validate routing before troubleshooting NAT.
- ✓ Use packet capture if needed.



KEY TAKEAWAY

A good understanding of NAT types and their use cases helps in effective troubleshooting and secure network design.

**Understand the traffic flow,
not just the translation!**



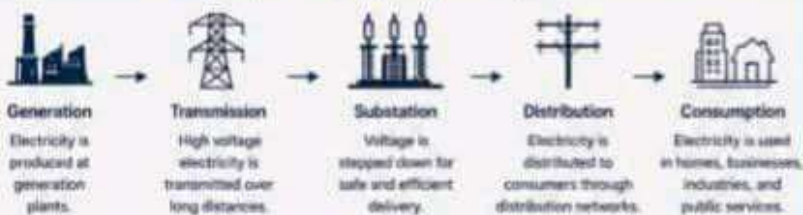
POWER SYSTEMS ENGINEERING

Delivering Reliable Energy for a Sustainable Future

Power systems engineering is the backbone of modern society. It ensures that electricity is generated, transmitted, distributed, and used efficiently and safely – 24/7.



WHAT IS A POWER SYSTEM?



KEY OBJECTIVES

- Ensure reliability and uninterrupted power supply
- Operate the system safely and efficiently
- Maintain power quality
- Minimize losses and operational costs
- Plan for future demand and integrate renewable energy

MAIN COMPONENTS OF A POWER SYSTEM



1. GENERATION

- Thermal (coal, gas, oil)
- Nuclear
- Hydroelectric
- Renewable (solar, wind, biomass)

Converts various forms of energy into electrical energy.



2. TRANSMISSION

- High Voltage (HV): 132 kV to 765 kV and above
- Overhead Lines
- Underground / Submarine Cables

Transports bulk power over long distances with minimal losses.



3. SUBSTATION

- Voltage Transformation
- Switching
- Protection
- Metering

Steps down voltage levels and ensures protection and control of the system.



4. DISTRIBUTION

- Medium Voltage (MV): 1 kV to 33 kV
- Low Voltage (LV): 120/240 V, 400/230 V

Distributes electricity to end-users through distribution feeders and service lines.



5. OPERATION & CONTROL

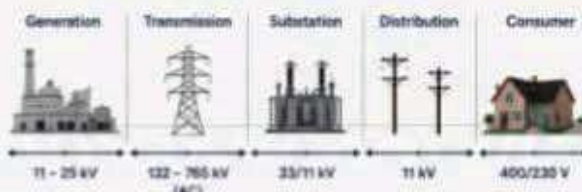
- System Monitoring (SCADA)
- Load Dispatch
- Fault Management
- Emergency Restoration

Monitors and controls the power system in real time to ensure reliability and security.

TYPICAL VOLTAGE LEVELS

System Part	Typical Voltage Range
Generation	6.6 kV – 25 kV
Transmission	132 kV – 765 kV and above
Sub-transmission	33 kV – 132 kV
Distribution (MV)	1 kV – 33 kV
Distribution (LV)	120/240 V, 400/230 V

POWER SYSTEM NETWORK (TYPICAL)



CHALLENGES IN POWER SYSTEMS

- Increasing demand for electricity
- Integration of renewable energy sources
- Aging infrastructure and asset management
- Maintaining stability and power quality
- Cybersecurity threats
- Extreme weather and climate change

POWER SYSTEM ENGINEERING DISCIPLINES



CAREER OPPORTUNITIES



TOOLS & SOFTWARE



POWERING TODAY.
EMPOWERING TOMORROW.

Building a smarter, cleaner,
and more reliable future together.





NETWORKING
WITH ISRAR

LEARN | PRACTICE | SUCCEED

BASIC ROUTER CONFIGURATION

STEP-BY-STEP GUIDE



Basic router configuration is the first step to make a router ready for network communication and remote management.

BASIC CONFIGURATION STEPS

-  **1** Enter Global Configuration Mode
Move from user EXEC mode to global configuration mode.
-  **2** Set Hostname
Assign a name to the router for easy identification.
-  **3** Disable DNS Lookup
Prevent the router from trying to resolve mistyped commands.
-  **4** Set Enable Secret Password
Secure privileged EXEC mode access.
-  **5** Set Console Password
Secure console access to the router.
-  **6** Set VTY Password
Secure remote access via Telnet/SSH.
-  **7** Save Configuration
Save the running configuration to NVRAM.

EXAMPLE: CONFIGURATION COMMANDS

```
Router> enable
Router# configure terminal
Router(config)# hostname R1                ! Set hostname
R1(config)# no ip domain-lookup            ! Disable DNS lookup
R1(config)# enable secret class123         ! Enable secret password
R1(config)# line console 0
R1(config-line)# password cisco123        ! Console password
R1(config-line)# login                     ! Enable login
R1(config-line)# exit
R1(config)# line vty 0 4
R1(config-line)# password cisco123        ! VTY password
R1(config-line)# login                     ! Enable login
R1(config-line)# transport input telnet ssh ! Allow Telnet & SSH
R1(config-line)# exit
R1(config)# end                            ! Return to privileged mode
R1# write memory                          ! Save configuration
R1#
```

SAMPLE TOPOLOGY



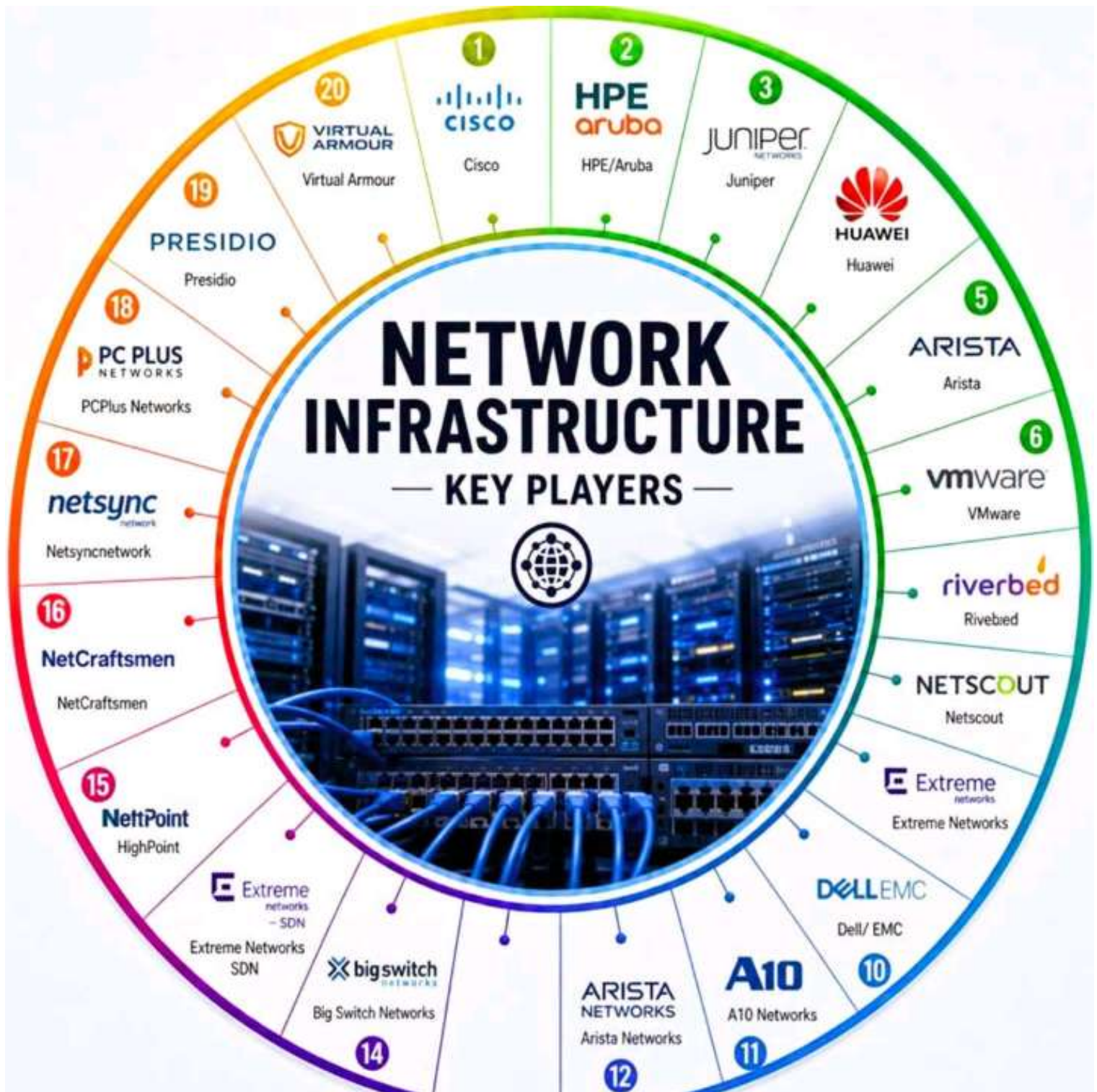
NOTE



This is the basic setup for any Cisco router. After this, you can configure interface IP addresses and routing as per your network requirements.

NETWORK INFRASTRUCTURE

— KEY PLAYERS —



SYSTEM SECURITY vs NETWORK SECURITY vs CYBER SECURITY

THREE AREAS. ONE GOAL: PROTECT, DETECT, RESPOND.



SYSTEM SECURITY

FOCUS

Protecting individual systems, servers, endpoints and the data they store from unauthorized access, malware and misuse.

RESPONSIBILITIES

- Secure operating systems (Windows, Linux)
- Manage user accounts and permissions
- Apply patches and updates
- Configure and harden systems
- Protect data on endpoints and servers
- Install and manage security software
- Monitor system logs and performance
- Backup and recovery
- Ensure system availability and integrity

SKILLS

- Windows / Linux Administration
- Active Directory
- File Permissions & Access Control
- System Hardening
- Virtualization
- Antivirus / EDR
- Backup & Recovery

TOOLS

- Windows Defender
- SCCM / WSUS
- VMware / Hyper-V
- Nessus / OpenVAS
- Acronis / Veeam
- Sysinternals Suite

WHERE USED?

On individual systems and servers (endpoints and local environments).

LEVEL

Entry to Mid Level



NETWORK SECURITY

FOCUS

Protecting the network infrastructure and the data traveling across it from unauthorized access, attacks and malicious activities.

RESPONSIBILITIES

- Design and implement secure network architecture
- Configure firewalls and VPNs
- Monitor network traffic and logs
- Detect and prevent intrusions
- Implement IDS/IPS solutions
- Secure wireless networks
- Segment networks and apply policies
- Manage network access control (NAC)
- Conduct vulnerability assessments

SKILLS

- Networking (TCP/IP, DNS, DHCP)
- Firewalls
- VPN & Encryption
- IDS / IPS
- Routing & Switching
- Network Monitoring
- Access Control (NAC)

TOOLS

- Cisco ASA / Firepower
- FortiGate / Palo Alto
- Wireshark
- Snort / Suricata
- SolarWinds / PRTG
- Nmap

WHERE USED?

Across the network infrastructure (routers, switches, firewalls, APs, etc.).

LEVEL

Mid Level



CYBER SECURITY

FOCUS

Protecting the organization as a whole from a wide range of cyber threats by people, processes and technology.

RESPONSIBILITIES

- Identify, analyze and respond to threats
- Perform penetration testing
- Conduct vulnerability assessments
- Manage security incidents
- Implement security policies and standards
- Security awareness and training
- Threat hunting and intelligence
- Forensics and malware analysis
- Ensure compliance and risk management

SKILLS

- Security Concepts
- Threat Analysis
- Penetration Testing
- Incident Response
- Digital Forensics
- Security Policies & Compliance
- Risk Management

TOOLS

- Kali Linux
- Burp Suite / OWASP ZAP
- SIEM (Splunk / ELK)
- Nessus / OpenVAS
- Metasploit
- TheHive / MISP

WHERE USED?

Across the entire organization (people, technology and processes).

LEVEL

Mid to Senior Level



KEY DIFFERENCES



Scope:
Focuses on individual systems and servers.

Goal:
Keep the system secure, stable and up to date.



Scope:
Focuses on the network infrastructure and traffic.

Goal:
Keep the network secure, available and performing.



Scope:
Focuses on the entire organization against all types of cyber threats.

Goal:
Protect the business, data and users from cyber risks.

CAREER PATH (EXAMPLE)



System Security Engineer



Network Security Engineer



Cyber Security Analyst



Senior Security Engineer



Security Architect / Manager



IN SHORT



SYSTEM SECURITY

You secure the system so it runs safely.



NETWORK SECURITY

You secure the network so data travels safely.



CYBER SECURITY

You secure the organization so everything stays safe.

★ All three are important and work together to build a strong security posture.

Learn. Practice. Get Certified. Stay Secure.





IPV4 SUBNETTING LAB

PRACTICE AND MASTER IPV4 SUBNETTING

OBJECTIVE

Practice IPv4 subnetting to create subnets, calculate network, broadcast, usable host ranges and understand CIDR notation.



TASKS

- 1 Subnet the given networks.
- 2 Find network, broadcast and usable host range.
- 3 Determine the number of usable hosts per subnet.
- 4 Write the CIDR notation and subnet mask.
- 5 Verify your results.



SUBNETTING REFERENCE

IPV4 ADDRESS CLASS

Class	Range (1st Octet)	Default Mask
A	1 – 126	255.0.0.0 (/8)
B	128 – 191	255.255.0.0 (/16)
C	192 – 223	255.255.255.0 (/24)
D	224 – 239	Multicast
E	240 – 255	Reserved

SUBNETTING FORMULA

- Number of Subnets = 2^n (n = borrowed bits)
- Number of Hosts per Subnet = $2^h - 2$ (h = host bits remaining)
- Network Address = All host bits = 0
- Broadcast Address = All host bits = 1
- Usable Host Range = Between Network+1 and Broadcast-1

CIDR / PREFIX

CIDR notation is written as /n where n is the number of 1s in the subnet mask.
Example: 192.168.10.0/26 → 26 bits are network bits.

PRACTICE PROBLEMS

#	Given Network	Prefix / Mask	Number of Subnets	Usable Hosts per Subnet	Network Address	First Usable Host	Last Usable Host	Broadcast Address	CIDR
1	192.168.10.0	/26	?	?	?	?	?	?	?
2	192.168.20.0	/27	?	?	?	?	?	?	?
3	10.0.0.0	/28	?	?	?	?	?	?	?
4	172.16.0.0	/25	?	?	?	?	?	?	?
5	192.168.1.0	/29	?	?	?	?	?	?	?

EXAMPLE SOLUTION (Row 1)

Given Network: 192.168.10.0/26
Subnet Mask: 255.255.255.192
Borrowed Bits: 2
Number of Subnets: $2^2 = 4$
Host Bits Remaining: 6
Usable Hosts per Subnet: $2^6 - 2 = 62$

Subnet #	Network	Host Range	Broadcast
1	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 – 192.168.10.254	192.168.10.255

SUBNETTING STEPS

- 1 Convert the subnet mask to binary.
- 2 Determine the number of borrowed bits.
- 3 Calculate number of subnets (2^n).
- 4 Calculate usable hosts per subnet ($2^h - 2$).
- 5 Find block size:

Block Size = 256 – Last Octet of Subnet Mask

- 6 List all subnets and calculate:

- ✓ Network Address (all host bits = 0)
- ✓ Broadcast Address (all host bits = 1)
- ✓ Usable Host Range (between them)

COMMON CIDR SHORTCUTS

/24 → 255.255.255.0 (254 hosts)
/25 → 255.255.255.128 (126 hosts)
/26 → 255.255.255.192 (62 hosts)
/27 → 255.255.255.224 (30 hosts)
/28 → 255.255.255.240 (14 hosts)
/29 → 255.255.255.248 (6 hosts)
/30 → 255.255.255.252 (2 hosts)



More borrowed bits = More subnets
Fewer host bits = Fewer usable hosts

QUICK TIPS

- ✓ Always write the mask in both decimal and CIDR.
- ✓ Use binary to avoid mistakes.
- ✓ Check the block size to jump between subnets.
- ✓ Double-check network and broadcast addresses.



CHALLENGE QUESTION

Given the network 192.168.50.0/27:

1. How many subnets and usable hosts per subnet are possible?
2. List all subnet network addresses.
3. What is the broadcast address of the last subnet?



Comment your answer below! 🗨️

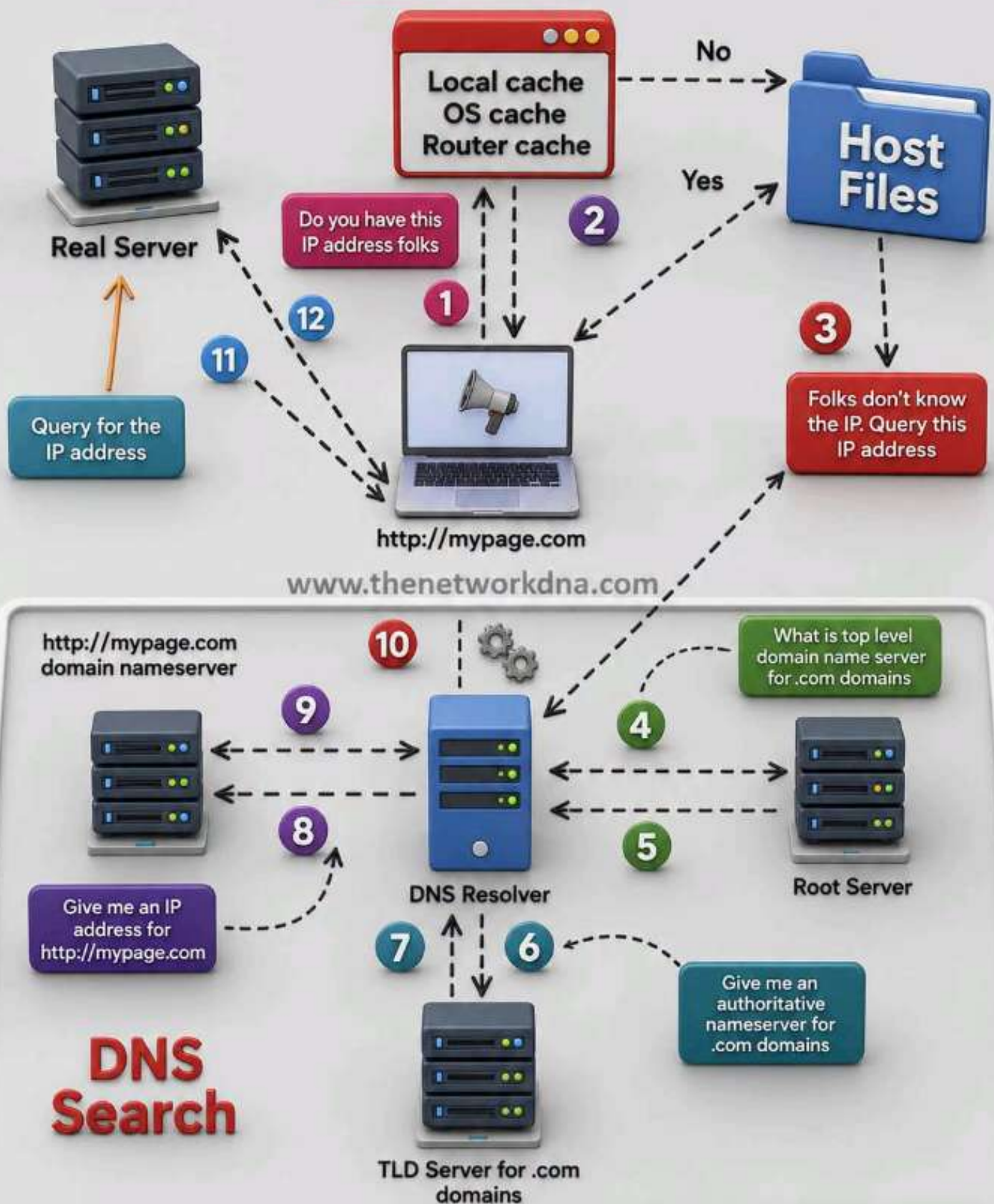


WHY THIS MATTERS

- ✓ Subnetting helps in efficient IP address management.
- ✓ Reduces broadcast traffic.
- ✓ Improves network performance and security.
- ✓ Essential skill for CCNA and network engineering roles.



HOW DNS WORKS



1. WELL-KNOWN PORT NUMBERS

PROTOCOL	PORT	TYPE
FTP (Data)	20	TCP
FTP (Data)	21	TCP
SSH	22	TCP
Telnet	23	TCP
SMTP	25	TCP
DNS	25	TCP/UDP
DHCP Server	67	UDP
DHCP Client	68	UDP
TFTP	69	UDP
HTTP	80	TCP
POP3	110	TCP
NTP	123	TCP
TMAP	143	TCP
SNMP	161	UDP
SNMP Trap	162	UDP
LDAP	389	TCP/UDP
HTTPS	443	TCP
SMB	445	TCP
Syslog	514	UDP
RIP	520	TCP
LDAPS	620	TCP
IMAPS	989/990	TCP
FTPS	993	TCP
POP3S	995	TCP
MS SQL	1433	TCP
Oracle	1521	TCP
L2TP	1701	TCP
RDP	3389	TCP
MySQL	3306	TCP
Postgre	5432	TCP
PostgreSQL	5900	TCP
VNC	5900	TCP
SIP	5066	TCP/UDP
SIP TLS	5061	TCP

2. ADMINISTRATIVE DISTANCE (AD)

PROTOCOL	AD VALUE
Connected	0
Static Route	1
EIGRP Summary	5
External BGP	20
Internal EIGRP	90
OSPF	110
IS - IS	115
RIP	120
External EIGRP	170
Internal BGP	200
Unknown	255

Sun Networking Academy

3. OSPF (OPEN SHORTEST PATH FIRST)

- Protocol Number : 89
- Multicast Address : 224.0.0.5
- DR/DDR Multicast : 224.0.0.6
- Default Cost Reference BW : 100 Mbps
- Router ID : 32-bit

OSPF Neighbor States:

1. Down
2. Init
3. 2-Way
4. Exstart
5. Exchange
6. Loading
7. Full

4. EIGRP (ENHANCED IGRP)

- Protocol Number : 88
- Multicast Address : 224.0.0.0
- AD (Internal) : 90
- AD (External) : 170

EIGRP K-VALUES (Default):

K1 = 1	K2 = 0	K3 = 1	K4 = 0	K5 = 0
--------	--------	--------	--------	--------

5. RIP (ROUTING INFORMATION PROTOCOL)

- Port : UDP 520
- Multicast : 224.0.0.9
- AD : 120
- Maximum Hop Count : 15
- Infinity : 16

6. BGP (BORDER GATEWAY PROTOCOL)

- Port : TCP 179
- EBGp AD : 20
- IBGP AD : 200

BGP Best Path Selection Order:

1. Weight (Cisco Specific)
2. Local Preference
3. Locally Originated
4. AS Path (Shortest)
5. Origin (Orig i < EGP < Incomplete)
6. MED (Lowest)
8. eBGP over iBGP
9. IGP Metric (Lowest)
10. Neighbor IP Address (Lowest)

Bus Topology

In **Bus Topology**, all devices are connected to a **single main cable** called the **backbone**.



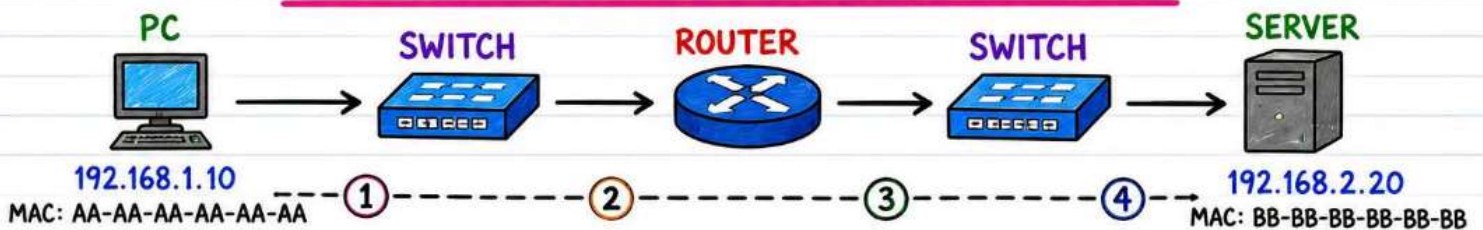
Key Point: All devices share the same cable. If the main cable fails, the whole network stops working.

Cybersecurity for Computer & AI HuB

FOLLOW FOR DAILY INSIGHTS

⌞ PACKET FLOW ⌞

PC → **SWITCH** → **ROUTER** → **SWITCH** → **SERVER**



★ HOW THE PACKET FLOW HAPPENS ★

① PC → SWITCH

- ★ PC checks the destination IP (192.168.2.20)
- ★ It sees the destination is in another network.
- ★ So, it sends the packet to Default Gateway (Router).
- ★ It uses ARP to find the Router's MAC.
- ★ Frame sent to SWITCH.

② SWITCH → ROUTER

- ★ Switch reads the destination MAC (Router's MAC).
- ★ It forwards the frame to the Router.

③ ROUTER → SWITCH

- ★ Router removes Layer 2 header (decapsulation).
- ★ Checks the destination IP (192.168.2.20) and finds the best path.
- ★ Router does ARP for Server's MAC.
- ★ Creates a new frame with new MAC addresses and forwards to SWITCH.

④ SWITCH → SERVER

- ★ Switch reads the destination MAC (Server's MAC).
- ★ It forwards the frame to the Server.
- ★ Server receives the packet. ✓

IMPORTANT

→ WHAT REMAINS SAME (End to End)

IP Addresses
(Source IP & Destination IP)



→ WHAT CHANGES AT EVERY HOP

MAC Addresses
(Source MAC & Destination MAC)

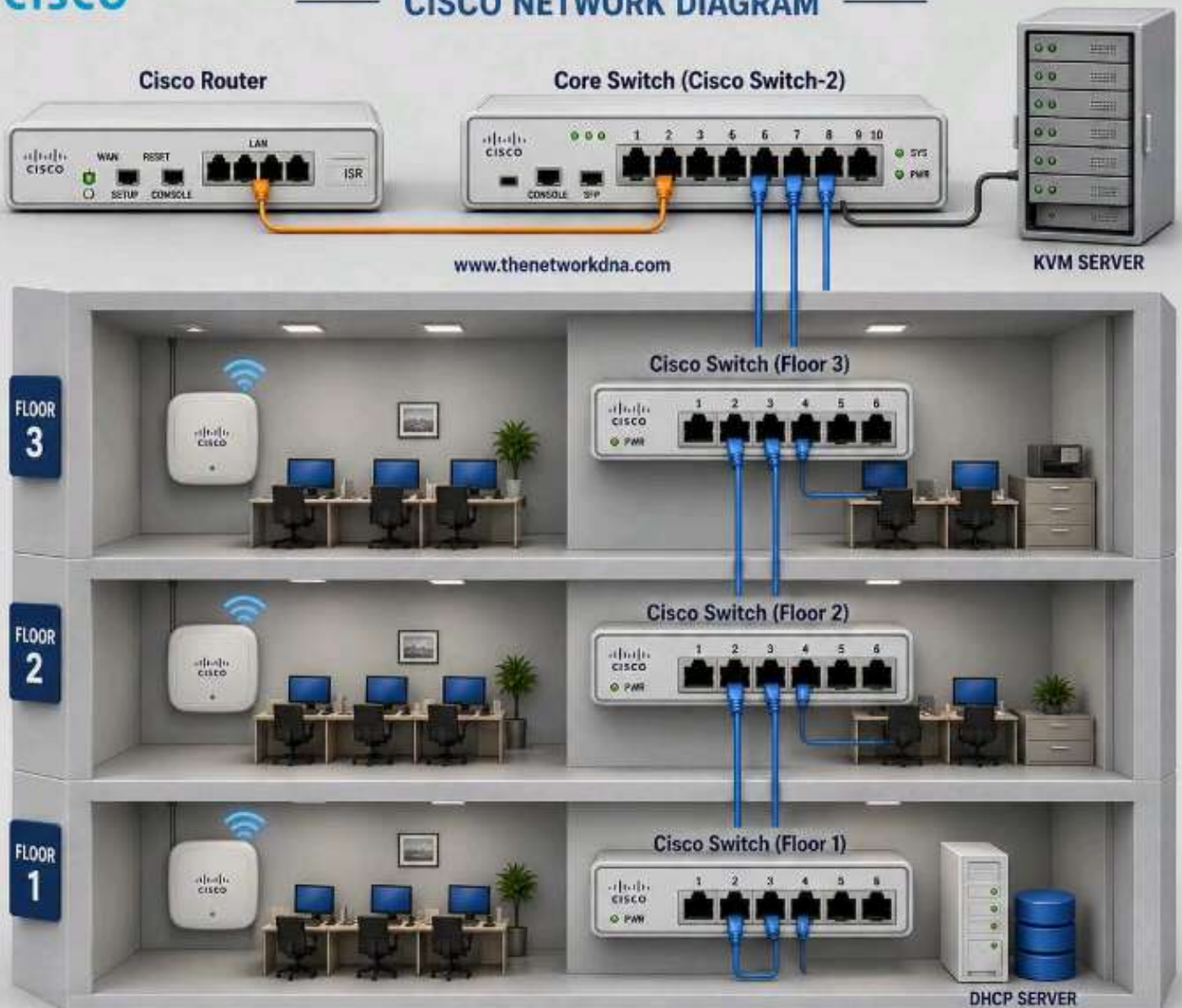


PM NETWORKING



SMALL OFFICE SETUP

CISCO NETWORK DIAGRAM



- Router connects to Core Switch
- Core Switch connects to Floor Switches
- Server connects to Core Switch

- ✓ Router only connects with Core Switch
- ✓ Core Switch connects with different Floor Switches
- ✓ Each Floor Switch connects to local devices (PCs, APs, Printers, etc.)

EIGRP vs OSPF COMPARISON LAB

COMPARE AND ANALYZE EIGRP AND OSPF IN A SAME NETWORK

OBJECTIVE

Compare EIGRP and OSPF in terms of routing table, convergence time, bandwidth usage and administrative distance.

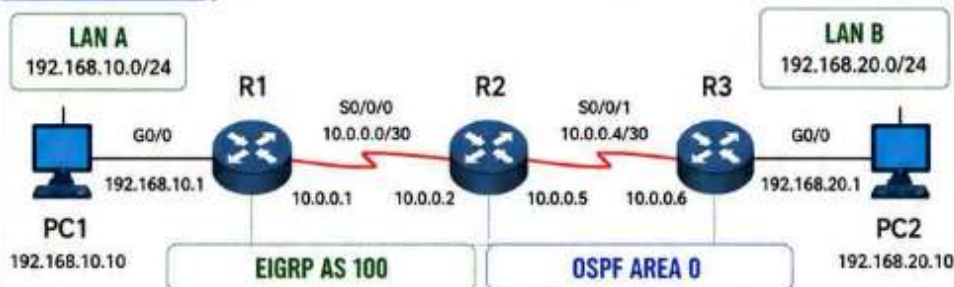


TASKS

1. Configure OSPF in Area 0.
2. Configure EIGRP AS 100.
3. Advertise all networks.
4. Compare routing tables.
5. Test convergence.
6. Compare administrative distance.
7. Analyze bandwidth usage.



TOPOLOGY



IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0	192.168.10.1	255.255.255.0	LAN A Gateway
R1	S0/0/0	10.0.0.1	255.255.255.252	To R2 (EIGRP)
R2	S0/0/0	10.0.0.2	255.255.255.252	To R1 (EIGRP)
R2	S0/0/1	10.0.0.5	255.255.255.252	To R3 (OSPF)
R3	S0/0/1	10.0.0.6	255.255.255.252	To R2 (OSPF)
R3	G0/0	192.168.20.1	255.255.255.0	LAN B Gateway
PC1	NIC	192.168.10.10	255.255.255.0	GW: 192.168.10.1
PC2	NIC	192.168.20.10	255.255.255.0	GW: 192.168.20.1

EIGRP CONFIGURATION (R1 & R2)

```

1 enable
2 conf t
3 router eigrp 100
4 no auto-summary
5 network 192.168.10.0 0.0.0.255
6 network 10.0.0.0 0.0.0.3
7 end
    
```

OSPF CONFIGURATION (R2 & R3)

```

1 enable
2 conf t
3 router ospf 1
4 network 10.0.0.4 0.0.0.3 area 0
5 network 192.168.20.0 0.0.0.255 area 0
6 end
    
```

VERIFICATION COMMANDS

- show ip route
- show ip protocols
- show ip eigrp neighbors
- show ip ospf neighbor
- show ip cef
- ping 192.168.20.10 (from PC1)
- ping 192.168.10.10 (from PC2)



These commands help you compare routing, convergence and protocol behavior.

EIGRP vs OSPF COMPARISON

Feature	EIGRP	OSPF
Type	Advanced Distance Vector	Link State
Update	Partial (Triggered)	Full (LSA based)
Metric	Bandwidth, Delay, Reliability, Load, MTU	Cost (Bandwidth)
Convergence	Very Fast	Fast
Administrative Distance	90 (Internal)	110 (Internal)
Classless Routing	Yes	Yes
VLSM / CIDR	Yes	Yes
Neighbor Discovery	Hello Packets	Hello Packets
Resource Usage	Lower	Higher (LSA flooding)
Best Use	Smaller to Medium Networks	Large & Scalable Networks

HOW TO TEST CONVERGENCE

1. Start a continuous ping from PC1 to PC2.
2. Shutdown the link between R2 and R3.
3. Note the time when ping starts failing.
4. Bring the link up again.
5. Note the time when ping starts succeeding.



Compare the recovery time (EIGRP vs OSPF).

CHALLENGE QUESTION



If the bandwidth of the link between R1 and R2 is reduced to 64 kbps, which protocol (EIGRP or OSPF) will adapt faster to the change? Why?

How will the routing table entries be different on R2 for both protocols?

Comment your answer below!

QUICK TEST

- ✓ What is the default AD of EIGRP?
- ✓ What type of routing is OSPF?
- ✓ Which protocol uses LSA?
- ✓ Which protocol has faster convergence?
- ✓ Which command shows routing protocols?





I turned my old phone
into a backup home
internet device, and it
saved my workday



Q Which devices operate only at the **Physical layer** of the Open Systems Interconnection (OSI) model? (Select two)



Hub



Repeater



Router



Switch

- A** Hubs.
- B** Repeaters.
- C** Routers.
- D** Switches.



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VLAN Configuration

– Cisco Switch –



• www.thenetworkdna.com •

01

CREATE
VLAN



1. Create a VLAN

```
Switch(config)# vlan 10
Switch(config-vlan)# name IT_Department
```



02

ASSIGN VLAN
TO A PORT



2. Assign VLAN to a Port

```
Switch(config)# interface Fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```



03

VERIFICATION



3. Verification

```
Switch# show vlan brief
Switch# show interface fa0/1 switchport
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2-24, Gi0/1-2
10	IT_Department	active	Fa0/1

04

VERIFICATION



4. Verification

```
Switch# show vlan brief
Switch# show interface fa0/1
switchport
```

```
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 10 (IT_Department)
Trunking VLANs Enabled: ALL
```



TIPS

- ✓ VLANs help segment your network for better security and performance.
- ✓ Access ports belong to one VLAN (untagged traffic).
- ✓ Use "show" commands to verify your configuration.



NETWORK SECURITY – CHEAT SHEET / ROADMAP

NETWORK FUNDAMENTALS

- OSI Model
- TCP/IP Model
- IP Addressing (IPv4/IPv6)
- Subnetting / CIBR
- VLANs, Trunks, .STP
- Routing Basics (Static, Default)
- Protocols: ARP, DNS, NCP, ICMP
- Common Ports (TCP/NS, SSH 22
 - HTTP 80, SNMP 161/142
 - RDP 3399, LDAP 389

ROUTING & SWITCHING

- Dynamic Dynamic: OSPF, EIGRP, GRP
- Policy Based Routing (PBB)
- Route Redistribution
- VRF, MPLS Basics (L3VPN)
- HSRP/VARP/GLBP
- EtherChannel (LACP)
- VLAN, VTP, PVST+, PSTF, MSTP

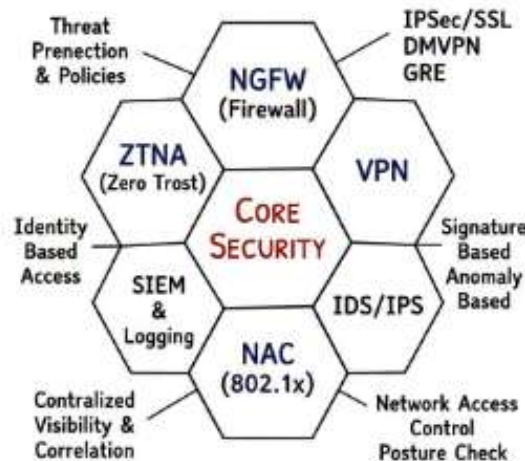
NETWORK SECURITY BASICS

- CIA Trad: Confidentiality, Integrity, Availability
- AAA: Authentication, Authorization, Accounting
- Access Control: ACLs
- Security Policies & Baselines
- Threats, Vulnerabilities, Exploits
- Defense in Depth

PERIMETER SECURITY

- Next-Generation Firewalls (NGFW)
- NAT, Static NAT
- DMZ Design
- VPN (Site-to-Site, Remote Access)
- Proxy, Content Filtering
- IDS / IPS

CORE SECURITY TECHNOLOGIES



SECURITY PROTOCOLS & SERVICES

- AAA: RADIUS, TACACS+
- 802.1X (Port-Based Access Control)
- NTP, SNMPv3, Spring
- SSH, NTTPS, SCP, SFTP
- IPsec (IKEv1 / IKEV2)
- TLS / SSL, DTLS
- PKI, Certificates, Digital Signatures

ADVANCED SECURITY

- Zero Trust Network Access (ZTNA)
- SASE (Secure Access Service Edge)
- NAC - Network Access Control
- Micro-segmentation
- DLP (Data Loss Prevention)
- SWG (Secure Web Gateway)
- CASB (Cloud Access Security Backlog)
- XDR / EDR / NDR
- Threat Intelligence

FIREWALL TECHNOLOGIES

- Palo Alto Networks (PAN-OS)
- Fortinet FortiGate (FortiNOS)
- Cisco Firepower (FTO)
- Cisco ASA (Legacy)
- Check Point
- Firewall Policies (Allow, Deny)
- NAT Policies, Security Zones
- Application Control, IPS, AV
- SSL Inspection, Decryption

AAA & ACCESS CONTROL

- TACACS+, RADIUS
- Role Based Access Control (RBAC)
- 802.1X / EAP-TLS
- Certificates (PKI)
- Cisco ISE / Aruba ClearPass

CLOUD NETWORKING & SECURITY

- AWS: VPC, Subnets, SG, NACL, TGM, Direct Connect, VPN, Route 53
- Azure: VNet, Azure Monitor
- GCP: VPC, Subnets, Firewall Rules, Cloud VPN, Cloud Router, Cloud NAT, Cloud DNS, Load Balancing
- Cloud Security Best Practices
- Hybrid Connectivity, Interconnect

MONITORING & LOGGING

- SIEM Platforms: Splunk, Microsoft Sentinel, QRadar
- Network Monitoring: Solarwinds, Log Analysis, Alerting
- NetFlow, eFlow
- Packet Capture: Wireshark, TCDump

COMPLIANCE & STANDARDS

- ISO 27001
- PCI-DSS
- NIST 800-SS
- GDPR

ROADMAP (STEP-BY-STEP)

Build Strong Networking Fundamentals
 Learn Routing & Switching in depth
 Understand Network Security Basics
 Learn Firewalls, VPNs, and Perimeter Security
 Master AAA, ACLs, and Security Protocols
 Hands-on Advanced Security Technologies
 Hands-on with Security Tools & Monitoring
 Learn Cloud & Modern Security Concepts
 Work on Real-world Scenarios / Labs
 Get Certified & Keep Learning

LAB IDEAS

- Build a small enterprise network in GNS3 / EVE-NG / Packet Tracer
- Configure Firewall, NAT, VPN
- Implement ACLs and 802.1X
- Set up SIEM Lab with Splunk (Free)
- Simulate Attacks & analyze logs

IMPORTANT CONCEPTS

- Least Privilege Access
- Always Verify, Never Trust
- Encrypt Everything
- Monitor Continuously
- Patch Regularly
- Backup & Test Recovery

USEFUL RESOURCES

- Cisco Networking Academy
- Palo Alto EBU-210
- Fortinet NSE Training
- TryHackMe / HackTheBox
- Wireshark University
- NIST Cybersecurity Framework
- SANS (Free Resource)
- YouTube: NetworkChuck, CBT Miggetts

KEY FORMULA

Subnet Notes = $2^n - 2$
 where n = (22 - Prefix Length)



BAKERY RAMAI TAPI LABA TIPIIS?

MUNGKIN 3 ATURAN INI BELUM KAMU TERAPKAN!

Kelola dengan cerdas, untung terus, bisnis pun bertumbuh.

1 HPP (FOOD COST) 30–45% dari Harga Jual Bersih

Harga jual bersih = Harga jual setelah dipotong biaya platform, promo, cashback, dll.

Contoh:

Harga jual di marketplace Rp50.000
Biaya platform & promo -Rp7.000

Harga jual bersih Rp43.000

Maka target HPP 30–45%
= Rp12.900 – 19.350



Jika HPP >45%

- ✗ Margin terlalu tipis
- ✗ Sulit berkembang
- ✗ Sedikit kenaikan harga bahan langsung mengurangi laba

Cara mencapainya

- ✓ Standarisasi resep
- ✓ Timbang semua bahan
- ✓ Minimalkan produk gagal
- ✓ Beli bahan baku dalam jumlah ekonomis
- ✓ Fokus pada produk dengan margin terbaik

2 BIAYA TENAGA KERJA Maksimal 18% dari Omzet

Contoh: Omzet Rp40.000.000/bulan

Target biaya tenaga kerja
≤ Rp7.200.000

Yang dihitung:



Jangan langsung merekrut jika:

- ✗ Oven masih kosong
- ✗ Masih banyak waktu menganggur
- ✗ Produksi belum optimal



3 BIAYA OPERASIONAL Maksimal 12% dari Omzet

Contoh: Omzet Rp40.000.000/bulan

Target operasional
≤ Rp4.800.000

Yang termasuk:



Catat semua pengeluaran operasional agar tidak bocor halus!



JIKA TARGET (RULE 2 DAN 3) BELUM TERCAPAI...

Jangan buru-buru naikan harga! Optimalkan produksi & manfaatkan sumber daya yang ada.

MAKSIMALKAN OVEN

Oven penuh, biaya per produk lebih rendah!



Oven penuh
Hemat energi,
HPP lebih rendah

VS



Oven hanya terisi 30%
Biaya hampir sama,
tapi hasil sedikit

PRODUKSI BATCH (1–2 kali seminggu)

Contoh:

Biasanya tiap hari
Mulai produksi
2x seminggu



Keuntungan:

- ✓ Gas lebih hemat
- ✓ Listrik lebih hemat
- ✓ Persiapan lebih singkat
- ✓ Tenaga lebih efisien

GUNAKAN FROZEN DOUGH

Adonan siap, panggang sesuai pesanan!



Produksi 100 adonan



Bekukan



Panggang
sesuai pesanan

Keuntungan:

- ✓ Waktu produksi lebih singkat
- ✓ Mengurangi lembur
- ✓ Stok siap
- ✓ Oven lebih efisien

GUNAKAN FROZEN CAKE

Panggang sekaligus, dekorasi sesuai order!



Panggang
sekali



Bekukan



Decor sesuai
order

Keuntungan:

- ✓ Kapasitas oven optimal
- ✓ Produk lebih konsisten
- ✓ Dekorasi lebih cepat

GUNAKAN TENAGA HARIAN

Fleksibel mengikuti naik turunnya pesanan!



Saat order
meningkat



Panggang tenaga
harian



Order turun,
tidak ada beban
gaji tetap

Lebih fleksibel & hemat biaya tetap!

OPTIMALKAN JADWAL PRODUKSI

Batch terjadwal = lebih rapi & efisien

Senin	Semua roti
Selasa	Cake
Rabu	Cookies
Kamis	Roti lagi
Jumat	Cake custom

Meminimalkan
waktu setup
& perpindahan
alat.



ILUSTRASI EFISIENSI

Produksi Harian

20 roti x 5 hari



= 100 roti

Produksi Batch

50 roti x 2 hari



= 100 roti

VS

- ✓ Hasil sama
- ✓ Oven dinyalakan lebih sedikit
- ✓ Waktu persiapan berkurang
- ✓ Biaya tenaga kerja lebih efisien

KERJA LEBIH RINGAN,
LABA LEBIH NYAMAN!



INDIKATOR BAKERY YANG SEHAT



HPP
30–45%



Upah & Gaji
≤18%



Operasional
≤12%



Masih ada laba
untuk investasi



Arus kas
positif



Tidak bergantung
pada utang
operasional

Bakery yang sehat bukan hanya menjual lebih banyak, tetapi mampu memanfaatkan oven, tenaga kerja, dan waktu produksi secara maksimal.

EFISIENSI ADALAH KUNCI!



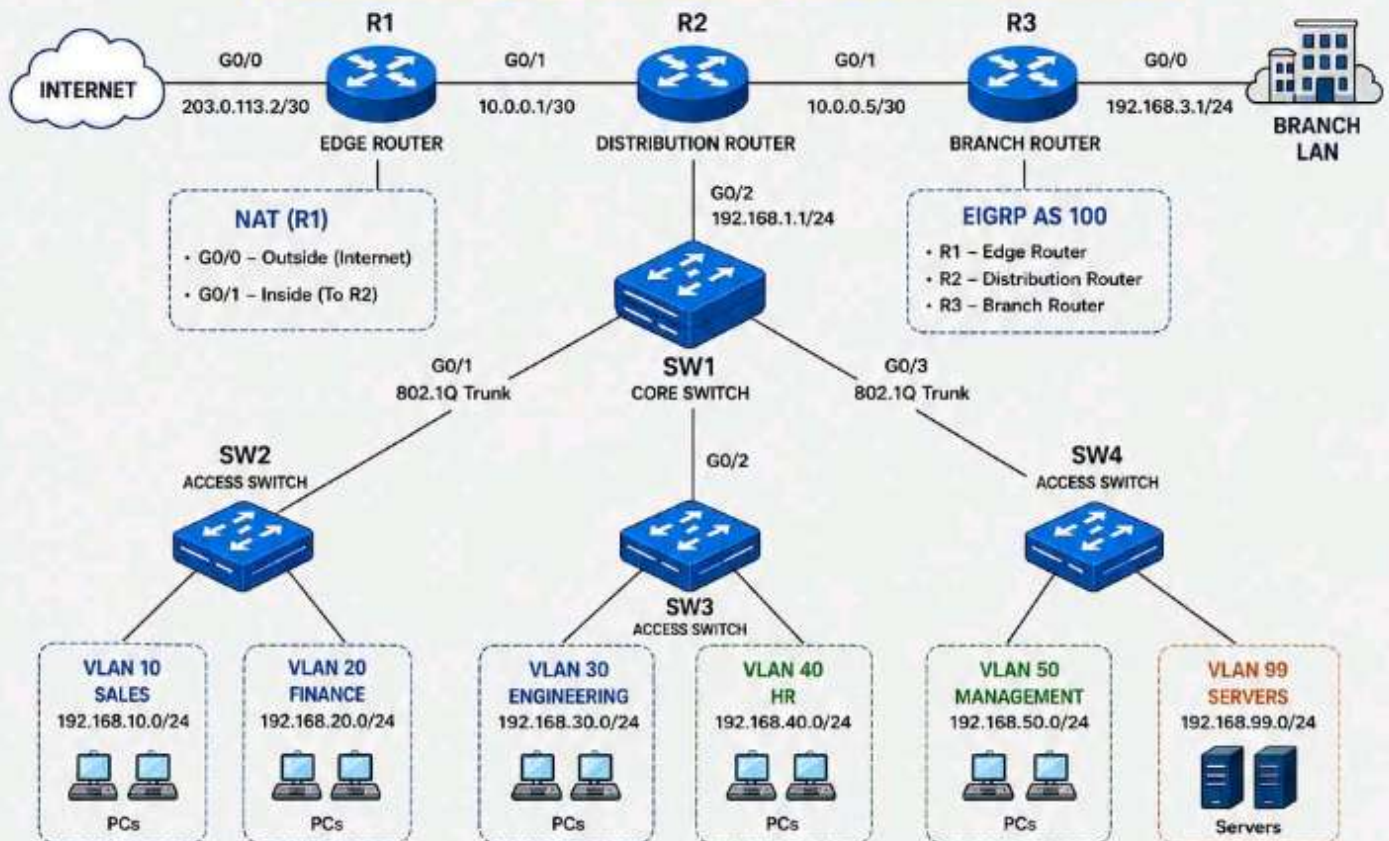
CATATAN PENTING

Persentase di atas merupakan target manajemen yang umum digunakan sebagai acuan. Angka ideal dapat berbeda tergantung jenis produk, model penjualan (offline/marketplace), skala usaha, dan strategi bisnis masing-masing bakery.



CCNA ENTERPRISE NETWORK CASE STUDY

VLANs + INTER-VLAN ROUTING + EIGRP + NAT + DHCP



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NETWORK OBJECTIVE



Departmental Segmentation using VLANs



Inter-VLAN Routing on Core Switch (SVIs)



Dynamic Routing with EIGRP



NAT on R1 for Internet Access



DHCP for End User VLANs



High Availability & Scalable Design

INTER-VLAN ROUTING (SW1 - SVIs)

VLAN	NETWORK	INTERFACE
10	192.168.10.0/24	G0/0.10
20	192.168.20.0/24	G0/0.20
30	192.168.30.0/24	G0/0.30
40	192.168.40.0/24	G0/0.40
50	192.168.50.0/24	G0/0.50
99	192.168.99.0/24	G0/0.99

EIGRP CONFIGURATION (AS 100)

R1:

- network 10.0.0.0 0.0.0.3
- network 203.0.113.0 0.0.0.3

R2:

- network 10.0.0.0 0.0.0.3
- network 192.168.1.0 0.0.0.255
- network 192.168.20.0 0.0.0.255
- network 192.168.30.0 0.0.0.255
- network 192.168.40.0 0.0.0.255
- network 192.168.50.0 0.0.0.255
- network 192.168.99.0 0.0.0.255

R3:

- network 10.0.0.4 0.0.0.3
- network 192.168.3.0 0.0.0.255

DHCP CONFIGURATION (R2)

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10

ip dhcp pool VLAN10_SALES
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8

|
(Repeat for other VLANs)
```

NAT CONFIGURATION (R1)

```
ip access-list standard NAT_INSIDE
permit 192.168.0.0 0.0.255.255

interface g0/0
ip nat outside

interface g0/1
ip nat inside

ip nat inside source list NAT_INSIDE
interface g0/0 overload
```

IP ADDRESS SUMMARY

SEGMENT	NETWORK	GATEWAY / INTERFACE	DESCRIPTION
R1 - R2 Link	10.0.0.0/30	R1: 10.0.0.1 R2: 10.0.0.2	WAN Link
R2 - R3 Link	10.0.0.4/30	R2: 10.0.0.5 R3: 10.0.0.6	WAN Link
R2 - Core LAN	192.168.1.0/24	R2 G0/2: 192.168.1.1	Core Management LAN
Branch LAN (R3)	192.168.3.0/24	R3 G0/0: 192.168.3.1	Branch Office LAN
VLAN 10 (SALES)	192.168.10.0/24	SW1 G0/0.10: 192.168.10.1	Users
VLAN 20 (FINANCE)	192.168.20.0/24	SW1 G0/0.20: 192.168.20.1	Users
VLAN 30 (ENGINEERING)	192.168.30.0/24	SW1 G0/0.30: 192.168.30.1	Users
VLAN 40 (HR)	192.168.40.0/24	SW1 G0/0.40: 192.168.40.1	Users
VLAN 50 (MANAGEMENT)	192.168.50.0/24	SW1 G0/0.50: 192.168.50.1	Users
VLAN 99 (SERVERS)	192.168.99.0/24	SW1 G0/0.99: 192.168.99.1	Servers



PRO TIP

Use VLANs for segmentation, EIGRP for intelligent routing, NAT for secure internet access, and DHCP for automation.



Secure



Scalable



Highly Available



Manageable



Cost Effective

LOCKOUT / TAGOUT (LOTO)



**ZERO ACCIDENTS
IS OUR GOAL**

LOCK IT. TAG IT. TEST IT. CONTROL IT.

A SIMPLE STEP TODAY CAN PREVENT A TRAGEDY TOMORROW.



WHAT IS LOTO?

Lockout/Tagout (LOTO) is a safety procedure used to ensure that hazardous energy sources are properly shut off and not re-energized before maintenance or servicing is performed on equipment.



**LOTO SAVES LIVES BY PREVENTING
UNEXPECTED ENERGIZATION.**



THE 5 STEPS OF EFFECTIVE LOTO

01

PREPARE



Understand the equipment, the hazards, and the energy sources.

02

SHUT DOWN



Shut down the equipment using the normal stopping procedure.

03

ISOLATE



Isolate all energy sources (electrical, mechanical, hydraulic, pneumatic, etc.).

04

LOCK & TAG



Apply lock and tag devices to ensure the energy source cannot be re-energized.

05

TEST & VERIFY



Test the equipment to verify zero energy state before starting the work.



NEVER REMOVE SOMEONE ELSE'S LOCK OR TAG.
ALWAYS FOLLOW YOUR COMPANY'S LOTO PROCEDURE.



WHY LOTO IS IMPORTANT?



**PREVENTS
SERIOUS INJURY
AND FATALITIES**



**PROTECTS
EQUIPMENT FROM
DAMAGE**



**ENSURES LEGAL
COMPLIANCE**



**IMPROVES
PRODUCTIVITY**



**BUILDS A STRONG
SAFETY CULTURE**

UPSKILL WITH WORLD-CLASS EHS COURSES



HSE COURSES

- IDHSE Course
- NEBOSH Courses
- IOSH Courses
- OSHA 30-Hour Course
- EOSH Course



ESC & QUALIFI

- ESC Level 6
- ESC Level 7
- Qualifi Level 6
- Qualifi Level 7



ISO LEAD AUDITOR COURSES

- ISO 45001 (OH&S)
- ISO 14001 (EMS)
- ISO 9001 (QMS)



FOOD SAFETY COURSES

ISO 22000:2018
Food Safety
Management System



AND MORE...

Industry-relevant
courses to build
competence and
advance your career.



OSI MODEL

OPEN SYSTEMS INTERCONNECTION

The OSI Model is a conceptual framework that standardizes network communication into **7 distinct layers**.



Remember: Data travels from **Layer 7** down to **Layer 1** on the sender side, and from **Layer 1** up to **Layer 7** on the receiver side.



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SYSTEM SECURITY vs NETWORK SECURITY vs CYBER SECURITY

THREE AREAS. ONE GOAL: PROTECT, DETECT, RESPOND.



SYSTEM SECURITY

FOCUS

Protecting individual systems, servers, endpoints and the data they store from unauthorized access, malware and misuse.

RESPONSIBILITIES

- Secure operating systems (Windows, Linux)
- Manage user accounts and permissions
- Apply patches and updates
- Configure and harden systems
- Protect data on endpoints and servers
- Install and manage security software
- Monitor system logs and performance
- Backup and recovery
- Ensure system availability and integrity

SKILLS

- Windows / Linux Administration
- Active Directory
- File Permissions & Access Control
- System Hardening
- Virtualization
- Antivirus / EDR
- Backup & Recovery

TOOLS

- Windows Defender
- SCCM / WSUS
- VMware / Hyper-V
- Nessus / OpenVAS
- Acronis / Veeam
- Sysinternals Suite

WHERE USED?

On individual systems and servers (endpoints and local environments).

LEVEL

Entry to Mid Level



NETWORK SECURITY

FOCUS

Protecting the network infrastructure and the data traveling across it from unauthorized access, attacks and malicious activities.

RESPONSIBILITIES

- Design and implement secure network architecture
- Configure firewalls and VPNs
- Monitor network traffic and logs
- Detect and prevent intrusions
- Implement IDS/IPS solutions
- Secure wireless networks
- Segment networks and apply policies
- Manage network access control (NAC)
- Conduct vulnerability assessments

SKILLS

- Networking (TCP/IP, DNS, DHCP)
- Firewalls
- VPN & Encryption
- IDS / IPS
- Routing & Switching
- Network Monitoring
- Access Control (NAC)

TOOLS

- Cisco ASA / Firepower
- FortiGate / Palo Alto
- Wireshark
- Snort / Suricata
- SolarWinds / PRTG
- Nmap

WHERE USED?

Across the network infrastructure (routers, switches, firewalls, APs, etc.).

LEVEL

Mid Level



CYBER SECURITY

FOCUS

Protecting the organization as a whole from a wide range of cyber threats by people, processes and technology.

RESPONSIBILITIES

- Identify, analyze and respond to threats
- Perform penetration testing
- Conduct vulnerability assessments
- Manage security incidents
- Implement security policies and standards
- Security awareness and training
- Threat hunting and intelligence
- Forensics and malware analysis
- Ensure compliance and risk management

SKILLS

- Security Concepts
- Threat Analysis
- Penetration Testing
- Incident Response
- Digital Forensics
- Security Policies & Compliance
- Risk Management

TOOLS

- Kali Linux
- Burp Suite / OWASP ZAP
- SIEM (Splunk / ELK)
- Nessus / OpenVAS
- Metasploit
- TheHive / MISP

WHERE USED?

Across the entire organization (people, technology and processes).

LEVEL

Mid to Senior Level



KEY DIFFERENCES



Scope:
Focuses on individual systems and servers.

Goal:
Keep the system secure, stable and up to date.



Scope:
Focuses on the network infrastructure and traffic.

Goal:
Keep the network secure, available and performing.



Scope:
Focuses on the entire organization against all types of cyber threats.

Goal:
Protect the business, data and users from cyber risks.

CAREER PATH (EXAMPLE)



System Security Engineer



Network Security Engineer



Cyber Security Analyst



Senior Security Engineer



Security Architect / Manager



IN SHORT



SYSTEM SECURITY

You secure the system so it runs safely.



NETWORK SECURITY

You secure the network so data travels safely.



CYBER SECURITY

You secure the organization so everything stays safe.

★ All three are important and work together to build a strong security posture.

Learn. Practice. Get Certified. Stay Secure.



What is DHCP?

Dynamic Host Configuration Protocol



Key Points:

- Automatically assigns IP addresses to devices.
- Eliminates manual configuration.
- Simplifies network management.



DHCP Client



DHCP Server

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The DORA Process:

1. **Discover:** Client broadcasts to find server.
2. **Offer:** Server offers an available IP.
3. **Request:** Client requests the offered IP.
4. **Acknowledge:** Server confirms the IP lease.



Key Terms:

- **Scope:** Range of available IP addresses.
- **Lease Time:** Duration a device keeps an IP.
- **Reservation:** Specific IP for a specific MAC.



Quick Facts:



Uses UDP Ports 67 & 68.



Prevents IP conflicts.



Works across subnets via Relay Agents.

Network Subnetting Simplify

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Top 50 Networking Protocol Interview

Questions with Answers (Part 2)

11. What is HTTPS?

Ans: Secure HTTP using SSL/TLS encryption.

Port: 443

Example: <https://google.com>

12. Difference between HTTP and HTTPS?

HTTP	HTTPS
Not encrypted	Encrypted
Port 80	Port 443
Less secure	More secure

13. What is FTP?

Ans: File Transfer Protocol transfers files.

Port: 21

Example: Uploading website files.

14. What is SFTP?

Ans: Secure File Transfer over SSH.

Port: 22

Example: Secure file upload.

15. What is TFTP?

Ans: Trivial File Transfer Protocol. Uses UDP.

Port: 69

Example: Router firmware update.

16. What is SMTP?

Ans: Simple Mail Transfer Protocol sends emails.

Port: 25

Example: Sending Gmail.

17. What is POP3?

Ans: Downloads email from server.

Port: 110

Example: Outlook downloading mail.

18. What is IMAP?

Ans: Keeps emails on the server.

Port: 143

Example: Accessing same inbox on phone and laptop.

19. Difference between POP3 and IMAP?

POP3	IMAP
Downloads emails	Synchronizes emails
Local storage	Server storage

20. What is DNS?

Ans: Domain Name System converts names to IP addresses.

Example: google.com

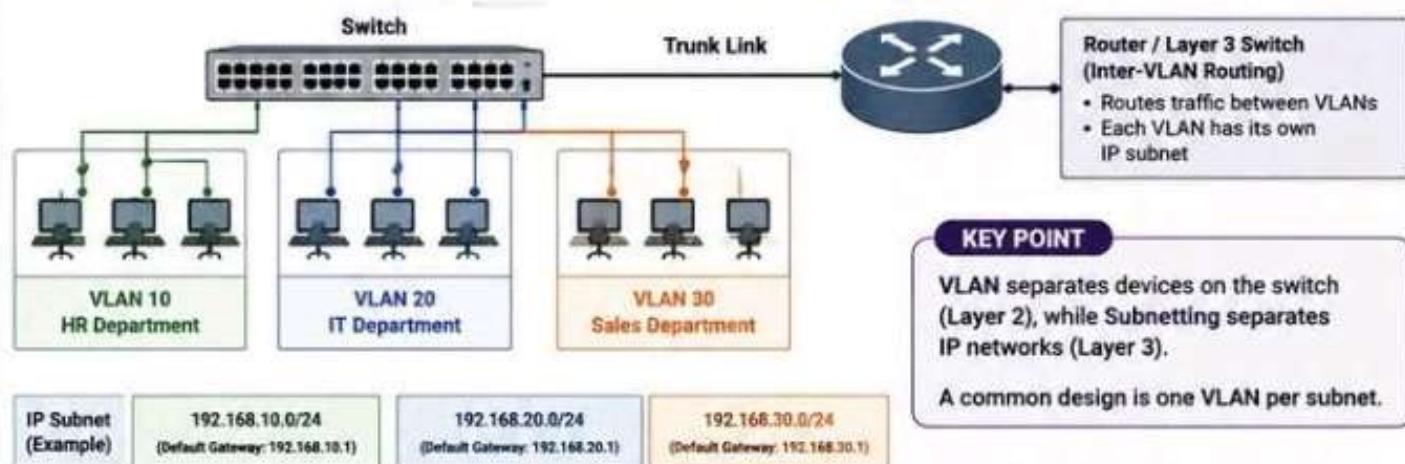
↓
142.250.xxx.xxx

SUBNETTING VS VLAN

Both are used to divide networks, but they work at different layers and have different purposes.

	SUBNETTING	VLAN (VIRTUAL LAN)
 DEFINITION	Subnetting is the process of dividing an IP network into smaller logical IP networks.	VLAN is the process of dividing a physical switch into multiple logical Layer 2 networks.
 LAYER	Works at Layer 3 (Network Layer).	Works at Layer 2 (Data Link Layer).
 IDENTIFICATION	Uses IP addresses and subnet masks .	Uses VLAN IDs (e.g., VLAN 10, VLAN 20).
 DEVICE USED	Requires a router or Layer 3 switch for communication between subnets.	Devices in different VLANs also need inter-VLAN routing (router or Layer 3 switch) to communicate.
 BROADCAST DOMAIN	Each subnet is a separate broadcast domain .	Each VLAN is a separate broadcast domain .
 MAIN GOAL	Better IP address management , efficient routing and scalability.	Improve security , reduce broadcasts, and separate departments/users.
 EXAMPLE	192.168.1.0/24 divided into: • 192.168.1.0/26 • 192.168.1.64/26 • 192.168.1.128/26 • 192.168.1.192/26	VLAN 10 (HR Department) VLAN 20 (IT Department) VLAN 30 (Sales Department)

HOW THEY WORK TOGETHER (EXAMPLE)



QUICK WAY TO REMEMBER

-  **VLAN** = Separates devices on a switch (Layer 2).
-  **Subnetting** = Separates IP address networks (Layer 3).

SUMMARY

- **Subnetting** uses IP addresses and subnet masks.
- **VLAN** uses VLAN IDs.
- Both are important for efficient, secure and scalable networks.

CCNA Routing and S...


NetworkDuck

Konten AI · 16 j ·

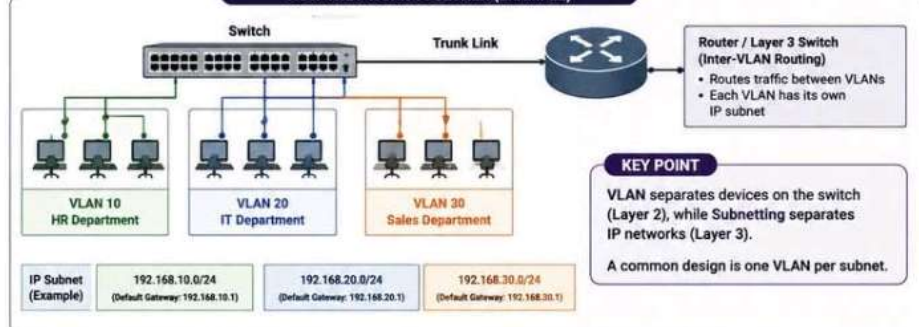
Subnetting vs vlan

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SUMMARY

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- VLAN uses VLAN IDs.
- Both are important for efficient, secure and scalable networks.



11


Edriskhan Halimi

16 j

Anda sedang offline.


EDR Technology

SUBNETTING

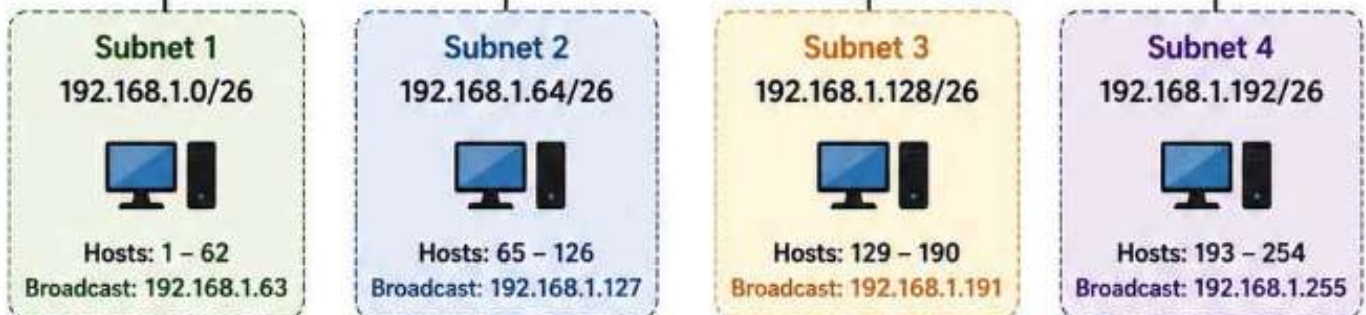
— Breaking a Network into Smaller Subnets —



Subnetting is the process of dividing a larger network into smaller, more efficient sub-networks. It improves IP address management, reduces broadcast traffic, and enhances network security.

EXAMPLE

Network: 192.168.1.0/24
(256 Addresses)



SUBNETTING STEPS

1

Identify the network and subnet mask.



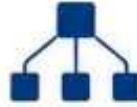
2

Determine the number of subnets needed.



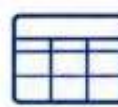
3

Borrow host bits to create the subnets.



4

Calculate the new subnet mask and block size.



5

Find the network, host range, and broadcast for each subnet.



SWITCH vs ROUTER vs HUB

Understand the Difference

SWITCH		ROUTER		HUB	
					
A switch connects devices within the same network using MAC addresses.		A router connects different networks together and routes data between them using IP addresses.		A hub connects devices in the same network and sends data to all connected devices.	
					
	Function		Function		Function
Connects devices within the same network (LAN)		Connects different networks and routes data between them		Connects devices within the same network (LAN)	
	Works At		Works At		Works At
Data Link Layer (Layer 2)		Network Layer (Layer 3)		Physical Layer (Layer 1)	
	Uses		Uses		Uses
MAC Addresses		IP Addresses		No MAC or IP awareness	
	Purpose		Purpose		Purpose
Forwards data to the specific device using MAC address		Finds the best path and routes data between different networks		Sends data to all ports (broadcasts)	
	Broadcast Domain		Broadcast Domain		Broadcast Domain
Each port is a separate collision domain		Separates broadcast domains		All ports in the same broadcast domain	
	Performance		Performance		Performance
High speed and more efficient		Slower than switch (depends on routing)		Low speed and less efficient	
	Typical Use		Typical Use		Typical Use
Connecting devices in homes, offices, or data centers		Connecting LAN to WAN, internet access, connecting multiple networks		Very small or old networks (rarely used now)	

QUICK SUMMARY



Switch

Best for connecting devices within the same network.



Router

Best for connecting multiple networks and the internet.



Hub

Outdated device that sends data to all devices.

